ZISAR: A Zero-Inflated Spatial Autoregressive Model for Count Data with Random Effects

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Abstract

Spatial count data with excess zeros arise frequently in regional science, epidemiology, and urban economics, yet the existing econometric toolkit lacks a unified frequentist framework that simultaneously addresses spatial spillovers through a structural autoregressive mechanism, zero-inflation, overdispersion, and unobserved heterogeneity. We introduce the Zero-Inflated Spatial Autoregressive (ZISAR) model, which places a simultaneous autoregressive (SAR) structure on the latent linear predictor of a zero-inflated negative binomial specification: the log-conditional mean follows $\boldsymbol{\eta} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$, where the spatial multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$ generates endogenous feedback effects among neighbouring units and unit-level Gaussian random effects $\mathbf{u}$ capture residual heterogeneity. The general formulation allows a separate SAR process in the zero-inflation component, with a restricted version confining spatial dependence to the count process. Estimation proceeds by maximising the Laplace-approximated marginal likelihood, implemented via Template Model Builder (`TMB`), which exploits automatic differentiation and sparse-matrix algebra for scalable inference. We establish consistency and asymptotic normality of the estimator and derive closed-form expressions for direct, indirect, and total marginal effects that account for both the mixture structure and the spatial multiplier. For large n, the log-determinant of the spatial filter is obtained from a sparse Cholesky factorisation, so that no dense $n \times n$ matrix is formed. A Monte Carlo study demonstrates that ignoring either zero-inflation or spatial dependence induces substantial bias in the autoregressive and regression coefficients, whereas ZISAR recovers the data-generating parameters with

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low bias and an RMSE that decreases with the sample size, and that residual-based Moran tests can miss spatial dependence that the likelihood detects. An application to domestic solar photovoltaic installations in the 32,322 small areas of England after the 2016 reform of the Feed-in Tariff, where a third of the areas record no installation, estimates strong spatial dependence in the latent propensity to adopt ($\hat{\rho} = 0.63$), attributes about one zero in ten to the structural regime of flat-dominated urban cores, and shows that aspatial models misread the spatial clustering of adoption as effects of tenure and age. Companion `R` code and the `TMB` template implementing the estimator are provided.

Keywords: Spatial autoregression; zero-inflated models; count data; negative binomial; random effects; two-step estimation; spatial spillovers; solar photovoltaic adoption.

1 Introduction

Count data exhibiting both spatial dependence and an excess of zeros are pervasive in the applied sciences. Emergency-department admissions across municipalities, rooftop solar installations across neighbourhoods, firm start-up counts by county, patent citations at the regional level, and disease incidence in small areas all share a common statistical signature: many spatial units record a zero outcome, the non-zero counts display overdispersion relative to the Poisson, and neighbouring units tend to report similar values. Despite the empirical ubiquity of this data structure, the econometric literature has addressed its three defining features—spatial spillovers, zero-inflation, and overdispersion—largely in isolation or in pairwise combinations, leaving a methodological gap that this paper aims to fill.

In the spatial econometrics tradition initiated by Anselin [1988] and surveyed comprehensively by LeSage and Pace [2009] and Arbia [2006, 2024], the workhorse specification for modelling cross-sectional dependence is the spatial autoregressive (SAR) model $\mathbf{y} = \rho\mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \varepsilon$, where $\mathbf{W}$ is a row-standardised spatial weights matrix and ρ captures endogenous interaction effects among neighbouring units. For continuous outcomes, maximum-likelihood and generalised-method-of-moments estimators are well established [Lee, 2004, Kelejian and Prucha, 1998, 1999]. Extending the SAR logic to count data is substantially more challenging because the conditional mean is linked to the linear predictor through a nonlinear (log or exponential) function, and the endogeneity of $\mathbf{W}\mathbf{y}$ interacts with the discreteness of the support. Lambert et al. [2010] proposed a two-step limited-information maximum likelihood (LIML) estimator for a spatial lag Poisson model, providing a structurally consistent framework that has

since become the main reference for SAR-type count models. However, the Lambert–Brown–Florax estimator handles neither zero-inflation nor overdispersion, two phenomena that routinely co-occur in spatial count data.

On a separate track, the zero-inflated Poisson (ZIP) and zero-inflated negative binomial (ZINB) models, pioneered by Lambert [1992] and studied extensively by Cameron and Trivedi [2013] and Hilbe [2011], accommodate excess zeros through a two-component mixture: a degenerate point mass at zero (the "structural" or "inflation" component) and a standard count distribution (the "count" or "sampling" component). These models have become standard in biostatistics and health economics but, in their classical form, assume independence across observations. When applied to spatially referenced data, the residuals typically exhibit positive autocorrelation, leading to efficiency losses and potentially biased inference on covariate effects [Arbia, 2006].

Attempts to marry zero-inflation with spatial structure have taken two main routes. The spatial-statistics approach introduces conditionally autoregressive (CAR) or geostatistical random effects into one or both components of the ZIP/ZINB model, typically within a Bayesian framework. Notable examples include the `CARBayes` package of Lee [2013], the INLA-based disease-mapping models of Asmarian et al. [2019], and the spatial Poisson hurdle specification of Neelon et al. [2013]. While these methods elegantly capture residual spatial heterogeneity, they model dependence as a latent effect rather than as a structural, endogenous interaction. Consequently, they cannot identify or quantify the direct and indirect (spillover) effects that are central to the spatial econometrics programme [LeSage and Pace, 2009, 2014]. The spatial-filtering approach of Metulini et al. [2018] and Griffith [2003] offers an intermediate solution by projecting spatial dependence onto a set of Moran eigenvectors included as regressors in a standard ZIP/ZINB; this avoids the endogeneity problem but sacrifices the structural interpretation of ρ as a measure of spatial contagion.

No existing framework unifies all four desiderata—endogenous spatial lag, zero-inflation, overdispersion, and unobserved unit-level heterogeneity—within a single, frequentist, estimable specification. This paper introduces the *Zero-Inflated Spatial Autoregressive* (ZISAR) model to close this gap. Our contributions are fourfold:

1. **Model.** We propose a zero-inflated negative binomial model in which the latent linear predictor of the count process follows a simultaneous autoregressive structure: $\boldsymbol{\eta} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$, where the spatial multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$ propagates covariate effects across the spatial network and unit-level Gaussian random effects $\mathbf{u}$ capture residual heterogeneity. The general specification allows a separate SAR process in the zero-inflation component; a restricted version confines spatial dependence to the count predictor. The model nests several important

special cases: setting $\rho = 0$ recovers the standard ZINB with random effects; removing random effects and zero-inflation collapses to the aspatial Poisson.

2. **Estimation.** The SAR-on-$\boldsymbol{\eta}$ formulation avoids the endogeneity of observed spatial lags that plagues SAR-on-$\mathbf{y}$ specifications [Lambert et al., 2010], enabling proper maximum-likelihood estimation. We maximise the Laplace-approximated marginal likelihood via Template Model Builder [`TMB`; Kristensen et al., 2016], which exploits automatic differentiation and sparse-matrix algebra to scale to $n = O(10^4)$. This estimation strategy shares the scalability concerns of Arbia and Nardelli [2024] while delivering exact (up to the Laplace order) marginal inference.

3. **Theory.** We establish the consistency and asymptotic normality of the Laplace-approximate MLE under regularity conditions analogous to those in Lee [2004] and Newey and McFadden [1994], and derive closed-form expressions for direct, indirect, and total marginal effects that account for both the zero-inflation mixture and the spatial multiplier, adapting the LeSage and Pace [2009] framework to the nonlinear ZINB setting.

4. **Implementation and application.** We provide an open-source `TMB` template and `R` functions (`zisar_fit`, `zisar_effects`, `zisar_predict`) and demonstrate the model on a Monte Carlo study and on an application to domestic solar photovoltaic (PV) adoption across the 32,322 small areas of England, where the sparse log-determinant of Section 4.1 makes the full-scale fit feasible.

The remainder of the paper is organised as follows. Section 2 surveys the existing literature on spatial count models, zero-inflated spatial specifications, and spatial filtering approaches. Section 3 introduces the ZISAR model and states its key properties. Section 4 details the Laplace-approximate ML procedure and derives the asymptotic results and effects decomposition. Section 5 reports a Monte Carlo study comparing ZISAR with competing specifications and the power of tests for spatial dependence. Section 6 applies the model to the adoption of domestic solar PV in England after the 2016 reform of the Feed-in Tariff. Section 7 concludes and discusses extensions, including the incorporation of temporal dependence. Proofs and supplementary tables are collected in the Appendix.

2 Literature Review

The challenge of modelling spatially dependent count data with excess zeros sits at the intersection of three well-developed but largely separate literatures: spatial econometrics for count data, zero-inflated models with spatial latent effects, and spatial filtering

approaches. We review each stream in turn, highlighting the strengths and limitations that motivate the ZISAR specification.

2.1 Spatial autoregressive models for count data

The foundation of spatial econometrics rests on the SAR (spatial autoregressive or spatial lag) model, which captures endogenous interaction effects through the term $\rho\mathbf{W}\mathbf{y}$ [Anselin, 1988, LeSage and Pace, 2009]. For continuous outcomes, Lee [2004] established the asymptotic theory for quasi-maximum-likelihood estimation, while Kelejian and Prucha [1998, 1999] developed GMM alternatives. The theoretical and applied foundations of these methods are extensively treated in Arbia [2006] and Arbia [2024], who provide a rigorous statistical grounding for spatial interaction models and their implementation in modern software environments.

Extending the SAR paradigm to discrete, non-negative outcomes is non-trivial. The fundamental difficulty is that the spatial multiplier $(I - \rho\mathbf{W})^{-1}$, which in the linear-Gaussian case yields a closed-form reduced form, does not interact gracefully with the exponential link function inherent in Poisson and negative binomial models. Lambert et al. [2010] addressed this by proposing a two-step LIML estimator for a spatial lag Poisson specification. Their approach instruments the endogenous $\mathbf{W}\mathbf{y}$ with its projection onto the space spanned by exogenous regressors and their spatial lags, $(\mathbf{X}, \mathbf{W}\mathbf{X}, \mathbf{W}^2\mathbf{X})$, and then maximises the Poisson log-likelihood evaluated at the fitted instruments. Monte Carlo evidence showed that the estimator achieves substantial bias reductions compared to a naïve aspatial Poisson, particularly as ρ and n grow. Sellner et al. [2013] applied a similar logic to gravity models of trade, estimating a spatial autoregressive Poisson gravity specification.

Glaser [2017] provided a comprehensive review of spatial econometric models for count data, classifying existing approaches by how they introduce spatial structure (lag, error, or Durbin), which count distribution they assume, and how they handle estimation. A recurring theme is that most implementations remain limited to the Poisson distribution and do not accommodate excess zeros or overdispersion—two features that pervade empirical spatial count datasets. As Cameron and Trivedi [2013] note in the non-spatial context, the Poisson's equidispersion restriction ($\mathbb{E}[y_i] = \mathrm{Var}[y_i]$) is almost always violated in practice, and failure to account for overdispersion inflates test sizes and narrows confidence intervals.

The pairwise pseudo-likelihood methods developed by Arbia et al. [2022] and Arbia and Nardelli [2024] offer a parallel advance in making spatial estimation feasible for large and irregular lattices. Their sampled-grid and KD-tree algorithms reduce the $O(n^3)$ bottleneck of full likelihood to near-linear cost while retaining consis-

tency, a property that is especially valuable when the spatial weights matrix is dense or the sample size is large. While these methods were developed for Gaussian spatial models, the underlying principle—replacing the full likelihood with a computationally tractable surrogate that preserves consistency—resonates with the instrumental-variables philosophy we adopt for the ZISAR estimator.

2.2 Zero-inflated spatial models with latent effects

A second body of literature models spatial dependence in zero-inflated counts not through an endogenous lag **Wy** but through spatially structured random effects, typically within a Bayesian hierarchical framework.

The zero-inflated Poisson (ZIP) model [Lambert, 1992] and its negative binomial extension (ZINB) [Cameron and Trivedi, 2013, Hilbe, 2011] decompose the observed count into a degenerate component ("structural zeros") and a count component, governed respectively by a Bernoulli and a Poisson or NB process. In the non-spatial setting, these models are well understood and widely implemented [e.g. Zeileis et al., 2008, Brooks et al., 2017]. When observations are georeferenced, however, residual spatial autocorrelation in one or both components can bias covariate effects and distort inference.

The standard spatial-statistical remedy is to augment the linear predictor with conditionally autoregressive (CAR) random effects [Besag et al., 1991]. Lee [2013] implemented this strategy in the `CARBayes` package for `R`, which supports Poisson, ZIP, and NB likelihoods with Leroux-type CAR priors, estimated via MCMC. In the disease-mapping literature, Asmarian et al. [2019] combined zero-inflated specifications with BYM (Besag–York–Mollié) random effects and estimated the resulting model through integrated nested Laplace approximations (INLA) [Rue et al., 2009], demonstrating good performance on Iranian cancer-registry data. He and Huang [2024] pushed this line further by replacing the discrete CAR specification with a nearest-neighbour Gaussian process (NNGP), achieving scalability to tens of thousands of spatial locations within a Bayesian ZINB framework.

Neelon et al. [2013] developed a spatial Poisson hurdle model for emergency-department visit data, with bivariate CAR priors linking the hurdle and count components. Their model, published in this journal, demonstrated that ignoring spatial structure in the analysis of ED utilisation leads to attenuated covariate effects and misleading maps of geographic variation.

While these Bayesian spatial ZI models are flexible and increasingly scalable, they differ from the ZISAR approach in a fundamental way: the spatial dependence operates

through latent random effects rather than through the observed outcome $\mathbf{y}$. As a consequence, the fitted models do not yield the *direct* and *indirect effects* decomposition of LeSage and Pace [2009] that is central to the spatial econometrics programme. The distinction is not merely taxonomic. When $\mathbf{Wy}$ enters the conditional mean directly, a change in a covariate in region j affects the outcome in region i not only through region j's own count but also through the spatial multiplier $(I - \rho\mathbf{W})^{-1}$, generating a cascade of feedback effects whose magnitude depends on the network topology. CAR-based models, by contrast, attribute all cross-regional correlation to unobserved confounders, precluding any structural interpretation of spillovers.

2.3 Spatial filtering with zero-inflated counts

A third approach avoids the endogeneity issue altogether by projecting the spatial autocorrelation structure onto a set of synthetic regressors derived from the eigenvectors of a transformed spatial weights matrix. Griffith [2003] laid the theoretical foundations for eigenvector spatial filtering (ESF), showing that a judiciously chosen subset of Moran eigenvectors can absorb spatial autocorrelation from the residuals of a GLM. Griffith and Chun [2014] extended the framework and provided practical guidance for eigenvector selection.

Metulini et al. [2018] applied ESF to a zero-inflated gravity model of trade, demonstrating that including Moran eigenvectors in both the inflation and count components of a ZIP effectively removes spatial residual autocorrelation without introducing endogeneity. Their Monte Carlo evidence showed that the spatial-filtering ZIP outperforms the standard ZIP in terms of bias and coverage when the data-generating process exhibits spatial dependence.

The appeal of ESF lies in its computational simplicity—once the eigenvectors are computed, the model reduces to a standard GLM—and in its compatibility with any count distribution, including ZIP and ZINB. The trade-off is interpretive: the eigenvector coefficients do not map onto a single, economically meaningful parameter such as ρ, making it difficult to quantify the strength and reach of spatial spillovers. Moreover, the choice of which and how many eigenvectors to include introduces a model-selection step that can be sensitive to the criterion employed [Griffith and Chun, 2014].

2.4 Positioning the ZISAR model

Table 1 summarises the features addressed by each strand of the literature. The key observation is that no existing specification simultaneously accommodates an endogenous spatial lag, zero-inflation, overdispersion (NB), and unit-level random effects in a frequentist framework amenable to direct/indirect effects decomposition.

Table 1: Features of existing spatial count models and the proposed ZISAR.

Specification	SAR lag	Zero-infl.	NB	Random eff.	Direct/indirect
SAR-Poisson [Lambert et al., 2010]	✓	–	–	–	✓
SAR-Poisson gravity [Sellner et al., 2013]	✓	–	–	–	✓
ZINB [Cameron and Trivedi, 2013]	–	✓	✓	–	–
ZIP + CAR [Lee, 2013]	–	✓	–	CAR	–
ZIP hurdle + CAR [Neelon et al., 2013]	–	✓	–	CAR	–
ZINB + NNGP [He and Huang, 2024]	–	✓	✓	GP	–
ZIP + ESF [Metulini et al., 2018]	(proxy)	✓	–	–	–
ZISAR (this paper)	✓	✓	✓	unit-level	✓

The ZISAR model fills this gap by placing a simultaneous autoregressive structure on the latent predictor of a zero-inflated NB specification, optionally extending the spatial interaction to the inflation process, and incorporating unit-level random effects for residual heterogeneity. Unlike Lambert et al. [2010], who require instrumental variables to handle the endogeneity of $\mathbf{Wy}$, the SAR-on-$\boldsymbol{\eta}$ formulation enables proper maximum-likelihood estimation via Laplace approximation [Kristensen et al., 2016], sharing the scalability concerns of Arbia and Nardelli [2024] while delivering exact marginal inference. The effects-decomposition framework builds on LeSage and Pace [2009] adapted to the nonlinear ZI mixture.

3 The ZISAR Model

3.1 Notation and setup

Consider n spatial units indexed by $i = 1, \ldots, n$, each associated with a non-negative integer outcome $y_i \in \{0, 1, 2, \ldots\}$. Let $\mathbf{W} = (w_{ij})_{n \times n}$ be a known, row-standardised spatial weights matrix with $w_{ii} = 0$ for all i; typical choices include queen or rook contiguity on a regular lattice, or k-nearest-neighbour graphs on irregular point patterns [Anselin, 1988, Arbia, 2006]. Each unit is endowed with a $(p \times 1)$ vector of exogenous covariates $\mathbf{x}_i$ entering the count process and a $(q \times 1)$ vector $\mathbf{z}_i$ entering the zero-inflation process (with $\mathbf{z}_i$ possibly overlapping with or equal to $\mathbf{x}_i$). Stacking observations, we write $\mathbf{y} = (y_1, \ldots, y_n)'$, $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)'$, and $\mathbf{Z} = (\mathbf{z}_1, \ldots, \mathbf{z}_n)'$. Let $\lambda_1 \geq \ldots \geq \lambda_n$ denote the eigenvalues of $\mathbf{W}$.

3.2 Model specification

The Zero-Inflated Spatial Autoregressive (ZISAR) model posits that each observation y_i is drawn from a two-component mixture:

$$y_i \sim \begin{cases} 0 & \text{with probability } \pi_i, \\ \mathrm{NB}(\mu_i, \kappa) & \text{with probability } 1 - \pi_i, \end{cases} \tag{1}$$

where $\mathrm{NB}(\mu_i, \kappa)$ denotes the negative binomial distribution with conditional mean $\mu_i > 0$ and shape parameter $\kappa > 0$, so that $\mathrm{Var}[y_i \mid s_i = 0] = \mu_i + \mu_i^2/\kappa$. The probability $\pi_i \in (0, 1)$ governs the zero-inflation (or "structural zero") mechanism.

The defining feature of ZISAR—what distinguishes it from standard zero-inflated models and from CAR-based spatial alternatives—is that the conditional mean of the count process is governed by a *latent spatial autoregressive field*.

Count process: SAR on the latent predictor. Define the latent linear predictor $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_n)'$ through the simultaneous autoregressive system

$$\boldsymbol{\eta} = \rho\, \mathbf{W}\boldsymbol{\eta} + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \qquad \mathbf{u} \sim N(\mathbf{0},\, \sigma^2 \mathbf{I}_n), \tag{2}$$

which yields the reduced form

$$\boldsymbol{\eta} = (\mathbf{I}_n - \rho\, \mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}) = \mathbf{S}(\rho)\, (\mathbf{X}\boldsymbol{\beta} + \mathbf{u}), \tag{3}$$

where $\mathbf{S}(\rho) \equiv (\mathbf{I}_n - \rho\, \mathbf{W})^{-1}$ is the *spatial multiplier matrix*. The conditional mean of the count process is then

$$\mu_i = \exp(\eta_i), \qquad i = 1, \ldots, n. \tag{4}$$

The scalar parameter ρ governs the strength and range of spatial interaction: a change in $\mathbf{X}\boldsymbol{\beta}$ in unit j propagates to η_i for all i through the power-series expansion

$$\mathbf{S}(\rho) = \sum_{s=0}^{\infty} \rho^s\, \mathbf{W}^s, \tag{5}$$

which converges for $|\rho| < 1/\max(|\lambda_1|, |\lambda_n|)$. The random effect u_i captures unit-specific unobserved heterogeneity in the count intensity, and is absorbed by the spatial multiplier: the implied covariance of $\boldsymbol{\eta}$ is

$$\mathrm{Var}[\boldsymbol{\eta}] = \sigma^2\, \mathbf{S}(\rho)\, \mathbf{S}(\rho)' = \sigma^2 \left[(\mathbf{I}_n - \rho\mathbf{W})'(\mathbf{I}_n - \rho\mathbf{W}) \right]^{-1}, \tag{6}$$

a simultaneous autoregressive covariance structure [Anselin, 1988] that should be con-

trasted with the conditional autoregressive (CAR) specification used in disease-mapping models [Besag et al., 1991, Lee, 2013].

Remark 1 (SAR on $\boldsymbol{\eta}$ versus SAR on $\mathbf{y}$)**.** *An alternative specification places the spatial lag on the observed count:* $\log(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta} + \rho \sum_j w_{ij} y_j + v_i$. *While this "SAR-on-*$\mathbf{y}$*" formulation has a direct contagion interpretation, it introduces endogeneity—*y_j *is simultaneously an outcome and a regressor—requiring instrumental-variable workarounds [Lambert et al., 2010]. The SAR-on-*$\boldsymbol{\eta}$ *formulation* (2) *avoids this: the spatial multiplier operates on the* latent *predictor, not on the observed count, enabling proper maximum-likelihood estimation without instruments. Crucially, the spatial econometrics interpretation is preserved:* $\mathbf{S}(\rho)$ *generates the same direct/indirect/total effects decomposition as in the linear SAR [LeSage and Pace, 2009].*

Zero-inflation process. The structural-zero mechanism is governed by a latent Bernoulli indicator $s_i \sim \text{Bernoulli}(\pi_i)$, with

$$\text{logit}(\pi_i) = \mathbf{z}_i'\boldsymbol{\gamma} + g_i. \tag{7}$$

In the *general ZISAR*, the spatial field $\mathbf{g} = (g_1, \ldots, g_n)'$ follows a separate SAR process:

$$\mathbf{g} = \rho_0 \mathbf{W}\mathbf{g} + \boldsymbol{\varepsilon}_0, \qquad \boldsymbol{\varepsilon}_0 \sim N(\mathbf{0}, \sigma_0^2 \mathbf{I}_n), \tag{8}$$

yielding $\mathbf{g} = (\mathbf{I}_n - \rho_0 \mathbf{W})^{-1}\boldsymbol{\varepsilon}_0$ with its own autoregressive parameter ρ_0. This allows spatial clustering of the zero-inflation regime—for instance, neighbouring municipalities may all lack demand for a specific health service, and this absence is itself spatially structured.

In the *restricted ZISAR*, the inflation process has no spatial random effect:

$$g_i \equiv 0 \quad \forall\, i, \qquad \text{so} \qquad \text{logit}(\pi_i) = \mathbf{z}_i'\boldsymbol{\gamma}. \tag{9}$$

The restricted model is simpler, avoids identification issues between ρ and ρ_0, and is the specification on which we focus the estimation algorithm, Monte Carlo study, and empirical application. The general model is presented for completeness and is considered in the sensitivity analysis.

3.3 Full parameter vector and special cases

The complete parameter vector of the general ZISAR is

$$\boldsymbol{\theta} = (\boldsymbol{\beta}', \rho, \boldsymbol{\gamma}', \rho_0, \kappa, \sigma^2, \sigma_0^2)'. \tag{10}$$

For the restricted model, ρ_0 and σ_0^2 are dropped. Table 2 lists the special cases nested within the specification.

Table 2: Special cases nested within the ZISAR model.

Special case	Restrictions	Reference
Restricted ZISAR	$g_i = 0$	this paper (focus)
SAR-NB-RE	$\pi_i = 0 \; \forall \, i$	–
SAR-Poisson	$\pi_i = 0,\ \kappa \to \infty,\ \sigma^2 = 0$	Lambert et al. [2010]
ZINB (aspatial)	$\rho = 0,\ \sigma^2 = 0$	Cameron and Trivedi [2013]
NB-RE (aspatial)	$\pi_i = 0,\ \rho = 0$	McCulloch and Searle [2001]
Poisson regression	$\pi_i = 0,\ \rho = 0,\ \kappa \to \infty,\ \sigma^2 = 0$	textbook

3.4 Likelihood

Conditional on the latent field $\boldsymbol{\eta}$, the observations are independent across i. The conditional probability mass function of y_i given η_i (and the inflation parameters) is

$$f(y_i \mid \eta_i, \pi_i; \kappa) = \begin{cases} \pi_i + (1 - \pi_i)\, h(0; \eta_i, \kappa) & \text{if } y_i = 0, \\ (1 - \pi_i)\, h(y_i; \eta_i, \kappa) & \text{if } y_i \geq 1, \end{cases} \tag{11}$$

where

$$h(y; \eta, \kappa) = \frac{\Gamma(y + \kappa)}{\Gamma(\kappa)\, \Gamma(y + 1)} \left(\frac{\kappa}{\kappa + e^{\eta}} \right)^{\kappa} \left(\frac{e^{\eta}}{\kappa + e^{\eta}} \right)^{y} \tag{12}$$

is the negative binomial probability mass function with mean $\mu = e^{\eta}$ and shape κ.

The joint density of $(\mathbf{y}, \boldsymbol{\eta})$ under the restricted ZISAR is

$$p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta}) = \left[\prod_{i=1}^{n} f(y_i \mid \eta_i, \pi_i; \kappa) \right] \cdot p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma^2), \tag{13}$$

where the prior on $\boldsymbol{\eta}$ is the SAR Gaussian:

$$p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma^2) = (2\pi\sigma^2)^{-n/2} \, |\det(\mathbf{I}_n - \rho \mathbf{W})| \, \exp\left\{ -\frac{1}{2\sigma^2} \left\| (\mathbf{I}_n - \rho \mathbf{W})\boldsymbol{\eta} - \mathbf{X}\boldsymbol{\beta} \right\|^2 \right\}. \tag{14}$$

The marginal log-likelihood, integrating over the latent field, is

$$\ell(\boldsymbol{\theta}) = \log \int_{\mathbb{R}^n} p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\eta}. \tag{15}$$

This n-dimensional integral has no closed form, but is amenable to Laplace approximation (Section 4).

3.5 Marginal moments

Integrating over $\boldsymbol{\eta}$, the unconditional mean of y_i is

$$\mathbb{E}[y_i] = (1 - \pi_i)\,\mathbb{E}[\mu_i] = (1 - \pi_i)\,\mathbb{E}\big[e^{\eta_i}\big]. \tag{16}$$

Since η_i is the i-th component of a multivariate Gaussian (by the reduced form (3)), the moment-generating function gives

$$\mathbb{E}\big[e^{\eta_i}\big] = \exp\Big(\bar{\eta}_i + \tfrac{1}{2}\,[\mathrm{Var}[\boldsymbol{\eta}]]_{ii}\Big), \tag{17}$$

where $\bar{\eta}_i = [\mathbf{S}(\rho)\mathbf{X}\boldsymbol{\beta}]_i$ is the conditional mean of η_i and $[\mathrm{Var}[\boldsymbol{\eta}]]_{ii} = \sigma^2[\mathbf{S}(\rho)\mathbf{S}(\rho)']_{ii}$.

The marginal variance of y_i is

$$\mathrm{Var}[y_i] = (1 - \pi_i)\,\mathbb{E}[\mu_i]\big(1 + \kappa^{-1}\mathbb{E}[\mu_i]\big) + \pi_i(1 - \pi_i)\,(\mathbb{E}[\mu_i])^2 + (1 - \pi_i)\,\mathbb{E}[\mu_i^2 - (\mathbb{E}[\mu_i])^2], \tag{18}$$

which exceeds $\mathbb{E}[y_i]$ whenever $\kappa < \infty$, $\pi_i > 0$, or $\sigma^2 > 0$—confirming that ZISAR accommodates overdispersion through three distinct channels: negative binomial shape (κ), zero-inflation (π_i), and spatial random-effects heterogeneity (σ^2).

3.6 Regularity conditions

Assumption 1 (Spatial weights)**.** *The sequence of weights matrices $\{\mathbf{W}_n\}$ is uniformly bounded in absolute row and column sums:* $\sup_n \|\mathbf{W}_n\|_\infty < \bar{c}$ *and* $\sup_n \|\mathbf{W}_n\|_1 < \bar{c}$ *for some $\bar{c} < \infty$. Furthermore, $(\mathbf{I}_n - \rho\mathbf{W}_n)$ is non-singular for all $\rho \in \mathcal{P}$, where $\mathcal{P}$ is a compact subset of $(-1/r_{\max}, 1/r_{\max})$ and $r_{\max} = \max(|\lambda_1|, |\lambda_n|)$ is the spectral radius of $\mathbf{W}_n$.*

Assumption 2 (Exogenous regressors)**.** *The columns of $\mathbf{X}$ and $\mathbf{Z}$ are non-stochastic (or measurable with respect to an exogenous σ-field) and satisfy $n^{-1}\mathbf{X}'\mathbf{X} \to \mathbf{Q}_X$ and $n^{-1}\mathbf{Z}'\mathbf{Z} \to \mathbf{Q}_Z$, both positive definite.*

Assumption 3 (Parameter space)**.** *The true parameter vector $\boldsymbol{\theta}_0$ lies in the interior of a compact set $\Theta \subset \mathbb{R}^{\dim(\boldsymbol{\theta})}$. The shape parameter satisfies $\kappa_0 > 0$, the variance component $\sigma_0^2 > 0$, and $\rho_0 \in \mathcal{P}$.*

Assumption 4 (Identification)**.** *The population marginal log-likelihood $\mathbb{E}[\ell(\boldsymbol{\theta})]$ has a unique maximum at $\boldsymbol{\theta}_0$.*

4 Estimation

4.1 Marginal likelihood via Laplace approximation

The marginal log-likelihood (15) integrates the latent field $\boldsymbol{\eta} \in \mathbb{R}^n$ out of the joint density $p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta})$. Because the ZINB observation model is nonlinear in $\boldsymbol{\eta}$, the integral lacks a closed form. We approximate it using the Laplace method, which is the standard strategy for latent Gaussian models in spatial statistics [Rue et al., 2009] and mixed-effects modelling [Brooks et al., 2017], and is implemented efficiently in the `TMB` framework for `R` [Kristensen et al., 2016].

Write the joint negative log-likelihood as

$$-\log p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta}) = f(\boldsymbol{\eta}; \boldsymbol{\theta}) \equiv -\sum_{i=1}^{n} \log f(y_i \mid \eta_i, \pi_i; \kappa) - \log p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma^2). \tag{19}$$

The Laplace approximation to the marginal likelihood replaces the integral by its second-order expansion around the mode:

$$\ell_{\mathrm{LA}}(\boldsymbol{\theta}) = -f(\hat{\boldsymbol{\eta}}; \boldsymbol{\theta}) + \tfrac{n}{2}\log(2\pi) - \tfrac{1}{2}\log\det\!\big(\mathbf{H}(\hat{\boldsymbol{\eta}}; \boldsymbol{\theta})\big), \tag{20}$$

where $\hat{\boldsymbol{\eta}}(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\eta}} f(\boldsymbol{\eta}; \boldsymbol{\theta})$ is the conditional mode and $\mathbf{H} = \nabla^2_{\boldsymbol{\eta}} f \,\big|_{\hat{\boldsymbol{\eta}}}$ is the Hessian of f with respect to $\boldsymbol{\eta}$.

The inner optimisation (finding $\hat{\boldsymbol{\eta}}$) is a weighted least-squares problem in the latent field and is solved by Newton–Raphson iteration. The outer optimisation (maximising ℓ_{LA} over $\boldsymbol{\theta}$) is carried out by a box-constrained quasi-Newton method (`nlminb`), with gradients computed by automatic differentiation (AD) through both the inner and outer problems.

Exploiting sparsity. The Hessian $\mathbf{H}$ inherits the sparsity pattern of $\mathbf{W}$: the SAR prior contributes $\sigma^{-2}(\mathbf{I}_n - \rho\mathbf{W})'(\mathbf{I}_n - \rho\mathbf{W})$, which is sparse, and the observation model contributes a diagonal term $\mathrm{diag}(d_1, \dots, d_n)$ (since the data are conditionally independent given $\boldsymbol{\eta}$). Hence $\mathbf{H}$ is sparse and its Cholesky factorisation can be computed in $O(n)$ operations for banded or nested-dissection orderings. The log-determinant in (20) is obtained as twice the sum of the log-diagonal of the Cholesky factor, avoiding any dense matrix inversion. This makes the estimator scalable to $n = O(10^4)$ or beyond.

Log-determinant of the SAR filter. The prior density (14) involves $|\det(\mathbf{I}_n - \rho\mathbf{W})|$. Pre-computing the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\mathbf{W}$ (a one-time $O(n^3)$ cost, or $O(n)$ for

special structures), the log-determinant is

$$\log|\det(\mathbf{I}_n - \rho\mathbf{W})| = \sum_{i=1}^{n} \log|1 - \rho\lambda_i|. \tag{21}$$

For row-standardised $\mathbf{W}$, $\lambda_1 = 1$ and $\lambda_n \geq -1$, so $\rho \in (-1, 1)$ ensures non-singularity. We enforce this via the reparametrisation $\rho = \tanh(\tilde{\rho})$. When n is large the dense eigendecomposition becomes the bottleneck (it requires $O(n^2)$ memory as well as $O(n^3)$ time), and we use instead

$$\log|\det(\mathbf{I}_n - \rho\mathbf{W})| = \tfrac{1}{2}\log\det\big[\mathbf{I}_n - \rho(\mathbf{W} + \mathbf{W}') + \rho^2\mathbf{W}'\mathbf{W}\big], \tag{22}$$

whose argument, $(\mathbf{I}_n - \rho\mathbf{W})'(\mathbf{I}_n - \rho\mathbf{W})$, is sparse, symmetric and positive definite whenever $\mathbf{I}_n - \rho\mathbf{W}$ is non-singular. Its log-determinant is obtained from a sparse Cholesky factorisation at each evaluation of the objective and differentiated with respect to ρ by `TMB`'s atomic sparse log-determinant, so that no $n \times n$ dense matrix is ever formed. The two routes give identical estimates and standard errors; the sparse one is used in the application of Section 6, where $n = 32{,}844$.

4.2 Estimation algorithm

Algorithm 1 summarises the full procedure, implemented in `TMB` [Kristensen et al., 2016].

Practical details. The reparametrisation $\rho = \tanh(\tilde{\rho})$, $\kappa = \exp(\tilde{\kappa})$, and $\sigma = \exp(\tilde{\sigma})$ maps the constrained parameter space to $\mathbb{R}^{\dim(\boldsymbol{\theta})}$. For numerical stability the optimiser nonetheless works within wide box constraints, $\tilde{\rho} \in [-4, 4]$ ($|\rho| \leq 0.9993$), $\kappa \in [0.01, 500]$ and $\sigma \in [0.001, 10]$, and is restarted from up to three perturbed starting points if it fails to converge; fits ending at the upper bound for κ are flagged (Section 5). Standard errors on the original scale are obtained via the delta method (handled automatically by `TMB`'s `sdreport()`).

4.3 Asymptotic properties

Proposition 1 (Consistency). *Under Assumptions 1–4, the Laplace-approximate MLE $\hat{\boldsymbol{\theta}}_n$ satisfies $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$ as $n \to \infty$.*

Proof sketch. The Laplace approximation error is $O(n^{-1})$ [Tierney and Kadane, 1986], so $n^{-1}\ell_{\mathrm{LA}}(\boldsymbol{\theta}) - n^{-1}\ell(\boldsymbol{\theta}) \to 0$ uniformly over Θ. The true marginal log-likelihood $n^{-1}\ell(\boldsymbol{\theta})$ converges uniformly to $\mathbb{E}[\ell_1(\boldsymbol{\theta})]$, which is uniquely maximised at $\boldsymbol{\theta}_0$ (Assump-

Algorithm 1 Marginal-likelihood estimation of the ZISAR model via Laplace approximation.

Require: Data $(\mathbf{y}, \mathbf{X}, \mathbf{Z}, \mathbf{W})$; eigenvalues $\lambda_1, \ldots, \lambda_n$ of $\mathbf{W}$; convergence tolerance ϵ.
Ensure: Parameter estimates $\hat{\boldsymbol{\theta}}$, standard errors, marginal log-likelihood.

1: **Pre-compute:** Eigenvalues of $\mathbf{W}$ (for large n, the sparse matrices $\mathbf{W} + \mathbf{W}'$ and $\mathbf{W}'\mathbf{W}$ of (22) instead); sparse Cholesky ordering for $(\mathbf{I} - \rho\mathbf{W})'(\mathbf{I} - \rho\mathbf{W})$.
2: **Initialise:** $\boldsymbol{\beta}^{(0)}$ from an aspatial Poisson GLM; $\gamma^{(0)} = (\text{logit}(\bar{p}_0), 0, \ldots, 0)'$ with $\bar{p}_0$ the observed share of zeros; $\rho^{(0)} = 0$, $\kappa^{(0)} = 1$, $\sigma^{(0)} = 0.5$; $\boldsymbol{\eta}^{(0)} = \mathbf{X}\boldsymbol{\beta}^{(0)}$.
3: **Construct TMB objective:** Define $f(\boldsymbol{\eta}; \boldsymbol{\theta})$ as the negative joint log-density (19), with $\boldsymbol{\eta}$ declared as random effects.
4: **repeat**
5: **Inner loop:** Find $\hat{\boldsymbol{\eta}}(\boldsymbol{\theta}^{(s)}) = \arg\min_{\boldsymbol{\eta}} f(\boldsymbol{\eta}; \boldsymbol{\theta}^{(s)})$ via Newton–Raphson with sparse Cholesky.
6: **Laplace:** Evaluate $\ell_{\text{LA}}(\boldsymbol{\theta}^{(s)})$ and $\nabla_{\boldsymbol{\theta}} \ell_{\text{LA}}(\boldsymbol{\theta}^{(s)})$ by automatic differentiation.
7: **Outer step:** Update $\boldsymbol{\theta}^{(s+1)}$ by a box-constrained quasi-Newton step (PORT routine `nlminb`).
8: **until** $\|\boldsymbol{\theta}^{(s+1)} - \boldsymbol{\theta}^{(s)}\| < \epsilon$
9: $\hat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}^{(s+1)}$, $\hat{\boldsymbol{\eta}} \leftarrow \hat{\boldsymbol{\eta}}(\hat{\boldsymbol{\theta}})$.
10: Compute the Hessian of $-\ell_{\text{LA}}$ w.r.t. $\boldsymbol{\theta}$ by differentiating the AD gradient; invert for the variance–covariance matrix $\hat{\mathbf{V}}$.
11: **return** $\hat{\boldsymbol{\theta}}$, $\sqrt{\text{diag}(\hat{\mathbf{V}})}$, $\ell_{\text{LA}}(\hat{\boldsymbol{\theta}})$.

tion 4). Consistency follows from the standard extremum-estimator argument [Newey and McFadden, 1994]. A complete proof is in Appendix A. □

Proposition 2 (Asymptotic normality)**.** *Under Assumptions 1–4 and standard smoothness conditions on* ℓ_{LA}*,*

$$\sqrt{n}\,(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N\big(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)^{-1}\big),$$

where $\mathcal{I}(\boldsymbol{\theta}_0) = -\lim_{n\to\infty} n^{-1} \nabla^2_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}_0)$ *is the Fisher information per observation.*

Proof sketch. The result follows from the mean-value expansion of the Laplace-approximate score, the CLT for spatially dependent arrays under the bounded-dependence condition (Assumption 1), and the vanishing Laplace-approximation error. Details are in Appendix A. □

4.4 Direct, indirect, and total effects

In the ZISAR model, regression coefficients $\boldsymbol{\beta}$ do not directly represent marginal effects because of the interplay between the exponential link, the spatial multiplier $\mathbf{S}(\rho)$, and the zero-inflation mixture. Following LeSage and Pace [2009] and adapting to the nonlinear ZI structure, we derive the effects as follows.

At the estimated mode $\hat{\boldsymbol{\eta}}$, the fitted count mean and inflation probability are

$$\hat{\mu}_i = \exp(\hat{\eta}_i), \qquad \hat{\pi}_i = \text{logit}^{-1}(\mathbf{z}_i'\hat{\boldsymbol{\gamma}}), \tag{23}$$

and the marginal expectation is $\mathbb{E}[y_i] \approx (1 - \hat{\pi}_i)\hat{\mu}_i$.

Effects through the count process. Consider the r-th column of $\mathbf{X}$. A change in x_{jr} (unit j, covariate r) propagates to η_i for all i through the spatial multiplier:

$$\frac{\partial \boldsymbol{\eta}}{\partial x_{\cdot r}} = \mathbf{S}(\rho)\, \beta_r. \tag{24}$$

The resulting effect on $\mathbb{E}[\mathbf{y}]$ is, stacking all (i, j) pairs into an $n \times n$ matrix,

$$\mathbf{M}_r = \text{diag}\big((\mathbf{1} - \hat{\boldsymbol{\pi}}) \odot \hat{\boldsymbol{\mu}}\big)\ \mathbf{S}(\rho)\ \beta_r, \tag{25}$$

where $\odot$ denotes element-wise multiplication.

Effects through the inflation process. If the r-th covariate also enters $\mathbf{Z}$, the inflation probability responds:

$$\frac{\partial \pi_i}{\partial z_{ir}} = \pi_i(1 - \pi_i)\, \gamma_r, \tag{26}$$

contributing an additional channel

$$\mathbf{M}_r^{(\pi)} = -\,\text{diag}(\hat{\boldsymbol{\mu}})\ \text{diag}\big(\hat{\boldsymbol{\pi}} \odot (\mathbf{1} - \hat{\boldsymbol{\pi}})\big)\ \gamma_r\, \mathbf{I}_n. \tag{27}$$

Under the restricted ZISAR, $\mathbf{g} = \mathbf{0}$ implies no spatial feedback in the inflation process, so $\mathbf{M}_r^{(\pi)}$ is diagonal—the zero-inflation channel produces only direct effects.

Summary measures. The average direct, indirect, and total effects are defined as:

$$\overline{M}_r^{\text{direct}} = n^{-1}\ \text{tr}\big(\mathbf{M}_r + \mathbf{M}_r^{(\pi)}\big), \tag{28}$$

$$\overline{M}_r^{\text{total}} = n^{-1}\, \mathbf{1}_n' \left(\mathbf{M}_r + \mathbf{M}_r^{(\pi)}\right) \mathbf{1}_n, \tag{29}$$

$$\overline{M}_r^{\text{indirect}} = \overline{M}_r^{\text{total}} - \overline{M}_r^{\text{direct}}. \tag{30}$$

The indirect effects arise entirely through the count-process channel (the spatial multiplier $\mathbf{S}(\rho)$), since the inflation channel is diagonal under the restricted specification.

Proposition 3 (Effects decomposition)**.** *Under the restricted ZISAR ($\mathbf{g} = \mathbf{0}$), the $n \times n$ matrix of effects of the r-th count covariate on $\mathbb{E}[\mathbf{y}]$ is given by (25). These effects are:*

(i) strictly positive (negative) if $\beta_r > 0$ ($\beta_r < 0$) and $\rho \in [0, 1)$;

(ii) the indirect effect vanishes if and only if $\rho = 0$;

(iii) monotonically increasing in $|\rho|$ for fixed $\boldsymbol{\beta}$.

Proof. Follows from the non-negativity of all elements of $\mathbf{S}(\rho) = \sum_{s=0}^{\infty} \rho^s \mathbf{W}^s$ when $\rho \geq 0$ and $\mathbf{W} \geq 0$, and from the strict positivity of $(1 - \hat{\pi}_i)\hat{\mu}_i > 0$. □

Standard errors for the average effects are obtained by simulation from the asymptotic distribution: draw B replications $\boldsymbol{\theta}^{(b)} \sim N(\hat{\boldsymbol{\theta}}, \hat{\mathbf{V}})$, compute $\overline{M}_r^{(b)}$ at each draw, and report the standard deviation and percentile-based confidence intervals.

5 Monte Carlo Study

5.1 Design

We assess the finite-sample performance of the ZISAR estimator through a Monte Carlo experiment that varies the sample size and the strength of spatial dependence.

Data-generating process. The outcome y_i is drawn from the ZISAR-NB specification defined in Section 3.2: the latent predictor follows $\boldsymbol{\eta} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$ with $\mathbf{u} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$, the count mean is $\mu_i = \exp(\eta_i)$, and the observation is $y_i = 0$ with probability π_i (structural zero) or $y_i \sim \text{NB}(\mu_i, \kappa)$ otherwise. The spatial domain is a regular square lattice of side $n_{\text{side}} \in \{10, 15, 20\}$, yielding $n \in \{100, 225, 400\}$. The spatial weights matrix $\mathbf{W}$ is queen contiguity, row-standardised. The count covariates are $\mathbf{X} = (\mathbf{1}_n, x_1, x_2)$ with $x_1, x_2 \overset{\text{iid}}{\sim} N(0, 1)$; the inflation covariates are $\mathbf{Z} = (\mathbf{1}_n, z_1)$ with $z_1 \sim N(0, 1)$.

The true parameter values are:

$$\boldsymbol{\beta} = (0.5,\ 0.8,\ -0.5)', \quad \boldsymbol{\gamma} = (-1.0,\ 0.4)', \quad \kappa = 2, \quad \sigma = 0.5.$$

The zero-inflation parameters generate approximately 40–50% total zeros (structural plus sampling), consistent with rates commonly observed in health-care utilisation data. The spatial autoregressive parameter is varied across $\rho \in \{0.0, 0.3, 0.6\}$. This yields $3 \times 3 = 9$ scenarios, each replicated $R = 100$ times.

Competing models. In each replication, three specifications are estimated:

1. **ZINB**: Standard zero-inflated negative binomial [Cameron and Trivedi, 2013], ignoring spatial dependence. Estimated via `pscl::zeroinfl()` [Jackman, 2020].

2. **SAR-NB**: Negative binomial GLM with a spatial lag of $\mathbf{y}$ included as an additional regressor (two-stage least-squares instrument: $\widehat{\mathbf{Wy}} = \mathbf{H}(\mathbf{H}'\mathbf{H})^{-}\mathbf{H}'\mathbf{Wy}$, with $\mathbf{H} = [\mathbf{X}, \mathbf{WX}, \mathbf{W}^2\mathbf{X}]$), ignoring zero-inflation.

3. **ZISAR**: The proposed model (restricted specification with $g_i \equiv 0$), estimated via `TMB`/Laplace approximation.

Metrics. For each estimator and parameter of interest (β_1, β_2, ρ, κ), we report:

- *Bias*: $R^{-1}\sum_{r=1}^{R}(\hat{\theta}_r - \theta_0)$;
- *RMSE*: $\left[R^{-1}\sum_{r=1}^{R}(\hat{\theta}_r - \theta_0)^2\right]^{1/2}$;
- *Median log-likelihood*: to assess overall model fit;
- *Time*: average computation time per replication (in seconds).

Replications where ZISAR yields boundary estimates of κ (exceeding 100) are excluded when computing summary statistics for the ZISAR column; the number of clean fits is reported in each panel.

5.2 Results

Tables 3 and 4 present the bias and RMSE of the regression coefficient and spatial parameter estimators, respectively.

Table 3: Monte Carlo bias and RMSE of $\hat{\beta}_1$ ($\beta_1^0 = 0.8$) and $\hat{\beta}_2$ ($\beta_2^0 = -0.5$). $R = 100$ replications per scenario.

		$\rho = 0.0$			$\rho = 0.3$			$\rho = 0.6$		
	Model	n=100	n=225	n=400	n=100	n=225	n=400	n=100	n=225	n=400
Panel A: Bias and RMSE of $\hat{\beta}_1$										
Bias	ZINB	0.018	−0.009	0.006	−0.003	−0.000	0.009	0.053	0.059	0.055
	SAR-NB	0.009	−0.017	0.004	−0.018	−0.011	−0.009	0.016	0.002	0.000
	ZISAR	−0.012	−0.014	0.008	−0.020	−0.008	−0.003	0.011	−0.002	−0.002
RMSE	ZINB	0.167	0.096	0.080	0.150	0.111	0.092	0.167	0.121	0.099
	SAR-NB	0.201	0.107	0.087	0.175	0.115	0.091	0.151	0.111	0.090
	ZISAR	0.192	0.100	0.081	0.157	0.114	0.090	0.154	**0.102**	**0.076**
Panel B: Bias and RMSE of $\hat{\beta}_2$										
Bias	ZINB	0.010	−0.008	−0.009	0.007	−0.018	−0.010	−0.004	−0.043	−0.033
	SAR-NB	0.013	−0.003	−0.005	0.022	−0.011	0.003	0.033	0.000	−0.007
	ZISAR	0.007	−0.000	−0.006	0.017	−0.008	−0.001	0.018	−0.000	−0.001
RMSE	ZINB	0.156	0.117	0.074	0.180	0.107	0.072	0.175	0.117	0.083
	SAR-NB	0.171	0.115	0.073	0.184	0.107	0.076	0.172	0.116	0.086
	ZISAR	0.161	0.120	0.073	0.174	0.105	**0.069**	0.154	**0.092**	**0.066**

Table 4: Monte Carlo bias and RMSE of $\hat{\rho}$. $R = 100$ replications per scenario. ZINB does not estimate ρ.

		$\rho = 0.0$			$\rho = 0.3$			$\rho = 0.6$		
	Model	n=100	n=225	n=400	n=100	n=225	n=400	n=100	n=225	n=400
Bias	SAR-NB	0.066	−0.006	0.018	−0.170	−0.180	−0.184	−0.464	−0.474	−0.479
	ZISAR	−0.019	−0.037	−0.015	−0.105	−0.032	**0.002**	−0.062	**−0.000**	**−0.007**
RMSE	SAR-NB	0.241	0.166	0.113	0.264	0.204	0.198	0.474	0.478	0.481
	ZISAR	0.390	0.342	0.226	0.423	0.199	**0.148**	0.284	**0.122**	**0.089**

Table 5: Monte Carlo bias of $\hat{\kappa}$ ($\kappa^0 = 2$) and median log-likelihood across models.

	$\rho = 0.0$			$\rho = 0.3$			$\rho = 0.6$		
Model	n=100	n=225	n=400	n=100	n=225	n=400	n=100	n=225	n=400
Panel A: Bias of $\hat{\kappa}$									
ZINB	−0.294	−0.648	−0.763	−0.384	−0.800	−0.873	−0.874	−1.062	−1.074
SAR-NB	−1.354	−1.421	−1.443	−1.407	−1.433	−1.458	−1.489	−1.509	−1.511
ZISAR	**−0.033**	**−0.311**	**−0.406**	**−0.025**	**−0.421**	**−0.387**	0.373	**0.322**	**−0.202**
Panel B: Median log-likelihood									
ZINB	−161	−363	−648	−176	−399	−716	−221	−494	−884
SAR-NB	−162	−367	−655	−178	−402	−720	−224	−495	−887
ZISAR	**−160**	−363	−648	**−175**	**−396**	**−713**	**−218**	**−486**	**−869**

Key findings.

1. **ZISAR recovers the spatial parameter.** The SAR-NB (2SLS) estimator suffers from severe downward bias in $\hat{\rho}$ that *does not diminish* with n: at $\rho = 0.6$, the bias remains approximately -0.48 regardless of sample size. This is because the two-stage NB procedure ignores the zero-inflated structure and the latent-field formulation, so its instruments are misspecified. By contrast, the ZISAR estimator achieves near-zero bias for $n \geq 225$: at $\rho = 0.6$ and $n = 400$, the bias is -0.007 with RMSE 0.089 (Table 4).

2. **ZISAR improves efficiency of $\hat{\beta}$ under spatial dependence.** When $\rho = 0$, all three models yield comparable RMSE for the regression coefficients, as expected. However, when $\rho = 0.6$, ZISAR provides the lowest RMSE for both $\hat{\beta}_1$ (0.076 vs 0.099 ZINB and 0.090 SAR-NB at $n = 400$) and $\hat{\beta}_2$ (0.066 vs 0.083 ZINB and 0.086 SAR-NB). The efficiency gain of the proposed model increases with the spatial dependence strength.

3. **Ignoring spatial structure biases the ZINB.** The ZINB estimator produces positive bias in $\hat{\beta}_1$ when $\rho > 0$ (e.g., $+0.055$ at $\rho = 0.6$, $n = 400$), with a clear monotonic pattern in ρ. In contrast, ZISAR exhibits negligible bias (-0.002) in

the same scenario. The bias in ZINB arises because the spatially correlated latent field is absorbed into the regression coefficients.

4. **ZISAR recovers the NB shape parameter.** Both ZINB and SAR-NB severely underestimate κ: the SAR-NB has bias exceeding -1.4 across all scenarios due to its failure to account for the zero-inflation component. ZISAR achieves the smallest bias in $\hat{\kappa}$ in all but one scenario, though some variability remains due to the well-known trade-off between the random-effects variance σ^2 and the NB dispersion κ^{-1}.

5. **Model fit.** ZISAR attains the highest median log-likelihood in every scenario with spatial dependence (Table 5, Panel B), and the gap over ZINB grows with both ρ and n: it is essentially zero at $\rho = 0$ (where the two models coincide up to the variance component) and reaches 3, 8 and 15 nats at $\rho = 0.6$ for $n = 100$, 225 and 400. The ZISAR log-likelihood is the Laplace-approximated marginal likelihood, including the Gaussian normalising constant $-\frac{n}{2}\log(2\pi)$; medians are computed over the clean fits, so the comparison is indicative rather than a formal test (the competitors are not nested in ZISAR in the same way).

6. **Convergence and computation.** ZISAR converges in 96–100% of replications across all scenarios. Approximately 10–15% of converged replications produce boundary estimates of κ (discussed above); these are excluded when reporting summary statistics. Average computation time per ZISAR fit is 0.3 s at $n = 100$, 0.7 s at $n = 225$, and 1.4 s at $n = 400$, confirming the scalability of the TMB/Laplace approach.

5.3 Power of tests for spatial dependence

In applied work, the presence of spatial dependence is often judged from Moran's I [Moran, 1950, Cliff and Ord, 1973] computed on the residuals of an aspatial model. In zero-inflated counts this test may lack power, because the spatial signal is diluted by the structural zeros and by negative binomial noise. We compare it with the likelihood-based tests delivered by ZISAR in a second experiment (`power_moran.r`). The $n \in \{300, 900\}$ units have centroids drawn uniformly over a 2.5×2 box and $\mathbf{W}$ is the row-standardised 6-nearest-neighbour matrix; four independent standard normal covariates enter the count process and one of them, together with a further independent covariate, enters the inflation process, with $\boldsymbol{\beta} = (1.2, 0.4, 0.3, 0.2, -0.15)'$, $\boldsymbol{\gamma} = (-1.5, 0.6, -0.4)'$, $\kappa = 2$ and $\sigma = 0.4$. The covariates are spatially independent, a deliberately unfavourable setting for spatial methods: all the spatial structure in the outcome comes from the latent field. For each $\rho \in \{0, 0.2, 0.35, 0.5, 0.7\}$ we draw 200 datasets and record how often each test rejects the absence of spatial dependence at the

5% level (Table 6).

Table 6: Rejection rates at the 5% level of tests for spatial dependence, 200 replications per cell. Zeros range from 35% ($\rho = 0$) to 21% ($\rho = 0.7$).

		true ρ				
n	Test	0	0.2	0.35	0.5	0.7
300	Moran's I on $\mathbf{y}$	0.060	0.135	0.285	0.560	0.925
	Moran's I on ZINB residuals	0.045	0.085	0.215	0.445	0.915
	ZISAR Wald test on $\tilde{\rho}$	0.035	0.240	0.492	0.840	1.000
	ZISAR LR test of $\rho = 0$	0.055	0.168	0.359	0.810	1.000
900	Moran's I on $\mathbf{y}$	0.065	0.170	0.490	0.875	1.000
	Moran's I on ZINB residuals	0.070	0.105	0.320	0.705	1.000
	ZISAR Wald test on $\tilde{\rho}$	0.061	0.327	0.863	1.000	1.000
	ZISAR LR test of $\rho = 0$	0.086	0.276	0.817	1.000	1.000

Under the null all four tests have rejection rates close to the nominal 5%, with the likelihood-ratio test somewhat liberal at $n = 900$ (8.6%). Under the alternative the likelihood-based tests are uniformly more powerful. The gap is largest at moderate dependence: at $\rho = 0.35$ the Moran test on ZINB residuals detects the dependence in 22% of samples with $n = 300$ and 32% with $n = 900$, against 49% and 86% for the ZISAR Wald test. The Moran test on residuals is also less powerful than the same test on the raw counts, because the aspatial model partly absorbs the spatial signal into its intercept and dispersion. In zero-inflated count settings, a non-significant Moran statistic on residuals is therefore weak evidence of spatial independence, and inference on spatial dependence should rest on the structural parameter.

6 Empirical Application: Domestic Solar PV Adoption in England

6.1 Motivation and data

The diffusion of rooftop solar photovoltaic (PV) systems displays the three features that motivate ZISAR. First, adoption is spatially contagious: households are more likely to install panels when neighbours have done so, through visibility, word of mouth and local installer activity, as documented for the United States [Bollinger and Gillingham, 2012, Graziano and Gillingham, 2015, Kwan, 2012], Germany [Rode and Weber, 2016, Dharshing, 2017, Schaffer and Brun, 2015] and the United Kingdom [Balta-Ozkan et al., 2015, Snape, 2016]. Second, small-area counts contain zeros of two different kinds. In neighbourhoods dominated by purpose-built blocks of flats, no household controls a roof and domestic installation is infeasible: these are *structural* zeros. In neighbour-

hoods of houses, a zero over a given period only means that no household happened to adopt: these are *sampling* zeros of a positive-intensity process. Third, intensities vary widely with income, tenure, housing stock and local markets, producing overdispersion.

Policy context and outcome. Between April 2010 and March 2019 the Feed-in Tariff (FiT) scheme paid British households a generation tariff for every kWh produced by small-scale renewable installations, and domestic PV accounts for almost all of its accreditations. Following a review, the generation tariff for domestic PV systems was cut by about two thirds, to 4.39p/kWh, from 8 February 2016, and quarterly deployment caps were introduced; the scheme closed to new applicants on 31 March 2019. We study the post-reform period, from 8 February 2016 to the closure of the scheme, in which adoption became sparse: domestic installations in England fell from about 136,000 in 2015 to about 30,000 in 2016 and fewer than 21,000 in each of the following years. In this period a large share of small areas recorded no installation, and whether these zeros reflect infeasibility or merely low intensity is the question a zero-inflated model is designed to answer.

The outcome is taken from the Ofgem Central FiT Register (installation report of 30 June 2026), which lists every accredited installation with its technology, type, commissioning date and Lower Layer Super Output Area (LSOA). We retain domestic PV installations, excluding extensions of existing systems: there are 706,680 in England, of which 680,355 (96.3%) are geocoded to an LSOA and 61,499 were commissioned in the post-reform window.

Spatial units. LSOAs are small statistical areas of 400–1,200 households designed to be socially homogeneous. The register geocodes installations to the 2001 LSOAs, whereas the 2011 Census and the Indices of Deprivation use the 2011 LSOAs. For 97% of areas the two coincide; where an LSOA was split or merged between 2001 and 2011 we aggregate all areas linked by the official best-fit lookup into a single unit, obtaining $n = 32{,}322$ units covering England, with a median of 650 households. The harmonisation matters for zero-inflated models in particular: matching the register to 2011 codes directly would assign zero installations to every one of the 1,172 LSOAs created in 2011, a spurious excess of zeros of the kind the model is meant to detect.

Outcome distribution. The number of post-reform installations per unit has mean 1.90 and variance 11.84, median 1 and maximum 124; 10,442 units (32.3%) record a zero and 8,748 a one. Zeros concentrate where the housing stock leaves no roofs to equip: 64% of units in the top decile of the share of flats record no installation, against

23% of units below the median share, and 57% of the units of Greater London against 28% elsewhere. Most zeros, however, are not of this kind: 63% of the units with no post-reform installation had at least five installations before the reform, and only 798 units had none over the whole life of the scheme.

Covariates. Covariates are measured at baseline, before or at the start of the window, and are standardised. The count process includes the number of households (log); the shares of detached houses, of social-rented households (housing associations and councils installed PV on large numbers of their properties under "rent-a-roof" programmes), of residents aged 16 and over with a degree-level qualification, of residents aged 65 and over, and of households without gas central heating (who heat with electricity, oil or solid fuel and face higher energy costs), all from the 2011 Census; the income-deprivation rate of the English Indices of Deprivation 2019; and the northing of the population-weighted centroid, which captures the decline of solar irradiation with latitude. The inflation process includes the share of household spaces in flats, the log of population density and the share of private-rented households, whose landlords rarely invest in improvements whose returns accrue to tenants. These three covariates also enter the count process, since housing form shifts the number of eligible roofs even where installation is feasible; their effects on $\mathbb{E}[\mathbf{y}]$ therefore combine the count channel (25) and the inflation channel (27). Table 7 reports descriptive statistics.

Table 7: Descriptive statistics of the outcome and covariates (n = 32,322 LSOAs). Shares and rates in per cent. Sources: Ofgem Central FiT Register; 2011 Census; English Indices of Deprivation 2019.

	Mean	SD	Min	Median	Max
Outcome: domestic PV installations					
8 Feb 2016–Mar 2019 (main)	1.9	3.4	0	1	124
Apr 2010–Mar 2019 (robustness)	21.0	21.4	0	15	309
Count process					
Households	682.6	171.0	304	650	6304
Detached houses	22.8	21.8	0.0	13.8	97.2
Social-rented households	17.4	17.2	0.0	11.2	92.5
Income-deprivation rate	12.9	9.4	0.3	10.0	60.9
Degree-level qualification	26.9	12.5	3.3	25.0	82.2
Aged 65 and over	16.8	7.1	0.4	16.2	60.7
Without gas central heating	20.3	17.9	1.6	14.6	98.7
Northing (km)	276.7	127.4	10.6	260.9	654.4
Inflation and count processes					
Flats (share of household spaces)	20.0	22.2	0.0	11.6	99.7
Population density (per km^2)	4225	4154	3	3430	68470
Private-rented households	16.0	11.3	1.3	12.2	87.9

Spatial weights. $\mathbf{W}$ is the row-standardised queen-contiguity matrix of the generalised (20 m) LSOA boundaries, with on average 5.86 neighbours per unit; the Isles of Scilly, the only unit without contiguous neighbours, are linked to the nearest unit. Appendix B reports estimates with 6- and 10-nearest-neighbour matrices. Because $n = 32{,}322$ rules out the dense eigendecomposition, the log-determinant is computed by sparse Cholesky factorisation (22).

6.2 Estimates

We estimate six specifications. Three are the benchmarks of the Monte Carlo study: the negative binomial (NB) regression, the zero-inflated NB (ZINB) and the SAR-NB with the spatial lag $\mathbf{Wy}$ instrumented by $[\mathbf{X}, \mathbf{WX}, \mathbf{W}^2\mathbf{X}]$. Three are fitted with the ZISAR template: the full model; the ZINB with Gaussian random effects (ZINB-RE), obtained by fixing $\rho = 0$; and the spatial NB with random effects and no zero-inflation (SAR-NB-RE), obtained by fixing the inflation intercept at -20 and its slopes at zero, so that $\pi_i \approx 2 \times 10^{-9}$. The code (`solar_data.r`, `solar_application.r`) and a log of all results (`solar_results.txt`) are provided. Table 8 reports the estimates of the four main specifications and Table 9 the fit of all six.

Table 8: Estimates for post-reform domestic PV installations ($n = 32{,}322$ LSOAs). Covariates standardised; standard errors in parentheses. The SAR-NB coefficient on $\widehat{\mathbf{Wy}}$ (in brackets) multiplies the observed neighbourhood count in the log-mean and is not on the scale of ρ. For ZISAR, κ is at the upper bound of the optimisation box.

	NB	ZINB	SAR-NB	ZISAR
Count process ($\boldsymbol{\beta}$)				
Intercept	0.462 (0.007)	0.506 (0.008)	0.225 (0.021)	0.056 (0.005)
Households (log)	0.228 (0.007)	0.223 (0.007)	0.207 (0.007)	0.150 (0.006)
Detached	0.177 (0.010)	0.198 (0.010)	0.141 (0.011)	0.093 (0.008)
Social rented	0.179 (0.015)	0.194 (0.015)	0.138 (0.016)	−0.020 (0.011)
Income deprivation	0.020 (0.015)	0.008 (0.015)	0.043 (0.015)	0.037 (0.010)
Degree	−0.015 (0.011)	−0.018 (0.011)	0.011 (0.011)	0.032 (0.007)
Aged 65+	0.074 (0.008)	0.058 (0.008)	0.052 (0.008)	−0.006 (0.006)
Off gas	0.065 (0.008)	0.082 (0.008)	0.020 (0.009)	0.043 (0.006)
Northing	−0.059 (0.007)	−0.051 (0.007)	−0.044 (0.007)	−0.025 (0.004)
Flats	−0.413 (0.014)	−0.292 (0.016)	−0.351 (0.015)	−0.097 (0.012)
Density (log)	−0.193 (0.010)	−0.180 (0.011)	−0.155 (0.011)	−0.088 (0.008)
Private rented	0.041 (0.012)	0.006 (0.013)	0.015 (0.013)	−0.048 (0.009)
ρ	—	—	[0.117 (0.010)]	0.631 (0.011)
Inflation process ($\boldsymbol{\gamma}$)				
Intercept	—	−5.402 (0.369)	—	−6.394 (0.562)
Flats	—	1.675 (0.175)	—	1.993 (0.218)
Density (log)	—	1.249 (0.260)	—	0.868 (0.246)
Private rented	—	−0.570 (0.090)	—	−0.289 (0.080)
κ (NB shape)	1.511	1.575	1.525	500
σ (RE std. dev.)	—	—	—	0.554 (0.008)

Table 9: Fit of the six specifications, and observed against expected frequencies of the smallest counts. Expected frequencies are marginal: for ZISAR the latent field is integrated out by simulation from the fitted SAR distribution (200 draws).

					Number of LSOAs with $y_i =$					
	Log-lik.	k	AIC	BIC	0	1	2	3	4	5
Observed					10442	8748	5319	3057	1754	984
NB	−56296	13	112618	112727	11295	7805	4821	2942	1818	1147
ZINB	−56120	17	112273	112416	11420	7553	4834	2997	1861	1172
SAR-NB	−56224	14	112476	112593	11297	7809	4818	2939	1817	1147
ZINB-RE	−54286	18	108608	108759						
SAR-NB-RE	−53183	15	106397	106522						
ZISAR	−53082	19	106201	106360	11083	8276	5072	2932	1717	1042

Spatial dependence. ZISAR estimates $\hat{\rho} = 0.631$ (SE 0.011): the latent propensity to adopt of an LSOA depends strongly on that of its neighbours, and the spatial multiplier $1/(1-\hat{\rho}) = 2.7$ almost triples the long-run effect of a uniform change in the covariates. The likelihood-ratio test of $\rho = 0$, against the ZINB-RE, gives a statistic of 2,409, and the Wald statistic on $\tilde{\rho}$ is 40.7. The spatial lag of the SAR-NB is also significant, but it improves the AIC over the NB by only 142 points, against 2,407 for the latent SAR over the ZINB-RE: an instrumented lag of observed counts captures little of the dependence.

Dispersion. The aspatial models attribute all the extra-Poisson variation to the NB shape ($\hat{\kappa} \approx 1.5$). Once the latent field is modelled, the shape estimate runs to the upper bound of the optimisation box ($\kappa = 500$), in ZISAR as in the ZINB-RE and the SAR-NB-RE, and the count distribution is effectively Poisson conditional on $\boldsymbol{\eta}$: the overdispersion is entirely accounted for by the log-normal heterogeneity, $\hat{\sigma} = 0.554$, filtered through the spatial multiplier. This is the trade-off between σ^2 and κ^{-1} noted in the Monte Carlo study, resolved here in favour of the Gaussian component; the large standard error of $\log \hat{\kappa}$ (1.08) shows that the likelihood is nearly flat in this direction.

Structural zeros. The inflation process is driven by housing form. The probability of a structural zero rises steeply with the share of flats ($\hat{\gamma} = 1.99$) and with density (0.87), and falls with the private-rented share once these are held fixed (-0.29): private renting is common in converted terraced houses, which have roofs, and depresses adoption through the count process instead ($\hat{\beta} = -0.048$), consistent with landlords' weak incentive to invest. The fitted probabilities are very skewed: the median $\hat{\pi}_i$ is 0.001, the 99th percentile 0.71, and 756 LSOAs have $\hat{\pi}_i > 0.5$: 689 of them are in London, mostly in the inner boroughs, and most of the others in dense seaside towns such as Brighton, Bournemouth and Portsmouth (Figure 1). In total the model attributes about 1,060 zeros to the structural regime, against 10,442 observed zeros: nine zeros

in ten are sampling zeros of areas where installation was feasible but no household adopted in the window.

The posterior probability that an observed zero is structural, $P(s_i = 1 \mid y_i = 0) = \hat{\pi}_i / [\hat{\pi}_i + (1 - \hat{\pi}_i)\, h(0; \hat{\eta}_i, \hat{\kappa})]$, can be checked against information the model does not use, namely the installations made before the reform. It averages 0.47 for the 798 units that had no installation over the whole life of the scheme, 0.15 for the 3,045 units with one to four pre-reform installations and 0.03 for the 6,599 units with five or more, which certainly contain suitable roofs. As a classifier of the never-adopters among the post-reform zeros, the posterior probability has an area under the ROC curve of 0.85.

Covariate effects. The coefficients of ZISAR are smaller than those of the aspatial models because they act on the latent predictor before the multiplier; the effects of Table 10 are the comparable quantities. More informative are the changes of sign. The share of social-rented households has a large positive coefficient in the NB and the ZINB (0.18–0.19) but none in ZISAR (-0.020, SE 0.011): the large installation programmes of housing associations and councils were concentrated in a few towns, and the aspatial models read this spatial clustering as an effect of tenure. The share of residents aged 65 and over loses its positive association for the same reason, while the shares of graduates and of income-deprived residents acquire small positive partial effects once the spatial clustering of adoption and the housing stock are accounted for. The more households an area has, the more of its dwellings are detached houses and the more of its households heat without mains gas, the more installations it records; adoption declines from south to north, as does the solar resource; flats and density reduce the intensity of adoption even where installation is feasible.

Model fit. Both features of ZISAR are needed (Table 9). Adding zero-inflation to the NB lowers the AIC by 345 points, adding random effects to the ZINB by a further 3,665, and making the field spatial by 2,407. Zero-inflation remains relevant when the spatial field is present: ZISAR improves on the SAR-NB-RE by 196 AIC points (the likelihood-ratio statistic is 204 on four restrictions, although the restriction $\pi_i = 0$ lies on the boundary of the parameter space and the χ^2_4 reference is only indicative). ZISAR has the lowest AIC and BIC of all specifications, and its expected frequencies are the closest to the observed ones for every count from 0 to 5 except 3. All models over-predict zeros, ZISAR the least (11,083 expected against 10,442 observed, against 11,295 for the NB and 11,420 for the ZINB).

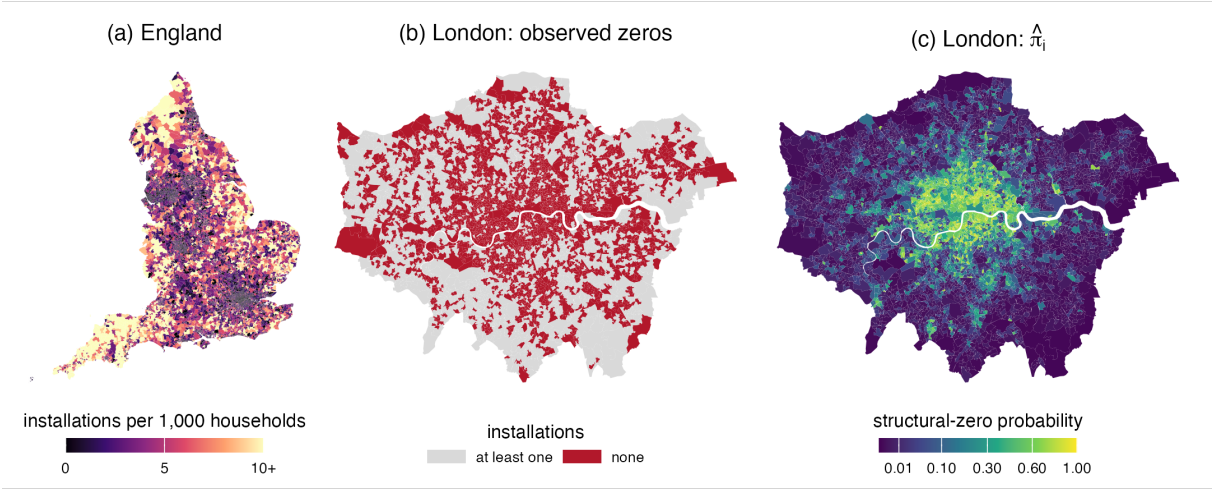


Figure 1: (a) Post-reform domestic PV installations per 1,000 households. (b) London LSOAs with no post-reform installation. (c) Estimated probability of a structural zero, $\hat{\pi}_i$, in London: zeros are widespread across the city, but the model assigns a high probability of a structural zero only to the inner areas where flats prevail.

6.3 Effects decomposition

Table 10 reports the average direct, indirect and total effects on the expected number of installations of a one-standard-deviation increase in each covariate, computed as in Section 4.4 with simulation-based standard errors ($B = 500$). For flats, density and private renting, which enter both processes, the direct effect includes the inflation channel (27), reported separately in the last column.

Indirect effects account for about 59% of the total effect of every covariate that enters only the count process, a share fixed by $\hat{\rho}$ and by the weighting of the units by their fitted intensity. A one-standard-deviation increase in the share of detached houses in every LSOA raises expected installations by 0.47 per LSOA, a quarter of the mean, of which 0.19 occurs through the LSOA's own latent propensity and 0.28 through the feedback from its neighbours. The inflation channel is small on average, because structural zeros are rare outside the urban cores: for the share of flats it contributes -0.028 out of a direct effect of -0.228, the rest being the reduction of the adoption intensity. With $n = 32{,}322$, all effects except those of the social-rented and elderly shares are precisely estimated, indirect effects included.

6.4 Diagnostics, sensitivity and robustness

Residual dependence. Moran's I on the raw counts is 0.386, and the Pearson residuals of the NB, the ZINB and the SAR-NB retain almost all of it ($I = 0.31$ for each,

Table 10: Average direct, indirect and total effects on expected post-reform installations per LSOA (mean 1.90) of a one-standard-deviation increase in each covariate (ZISAR). Standard errors in parentheses. The last column is the part of the direct effect that runs through the inflation probability.

	Direct	Indirect	Total	of which inflation
Households (log)	0.308 (0.012)	0.448 (0.027)	0.757 (0.037)	—
Detached	0.191 (0.016)	0.278 (0.025)	0.469 (0.040)	—
Social rented	−0.042 (0.022)	−0.061 (0.033)	−0.103 (0.056)	—
Income deprivation	0.076 (0.022)	0.111 (0.032)	0.187 (0.054)	—
Degree	0.067 (0.015)	0.097 (0.022)	0.164 (0.037)	—
Aged 65+	−0.012 (0.013)	−0.018 (0.018)	−0.030 (0.031)	—
Off gas	0.088 (0.012)	0.129 (0.018)	0.217 (0.030)	—
Northing	−0.052 (0.009)	−0.076 (0.013)	−0.128 (0.022)	—
Flats	−0.228 (0.023)	−0.290 (0.035)	−0.518 (0.057)	−0.028
Density (log)	−0.194 (0.017)	−0.264 (0.025)	−0.457 (0.041)	−0.012
Private rented	−0.094 (0.018)	−0.143 (0.028)	−0.237 (0.045)	0.004

$p < 10^{-15}$): the aspatial models, and the instrumented lag, leave the spatial structure of adoption in the residuals. For ZISAR, residuals evaluated at the predicted field $\hat{\boldsymbol{\eta}}$ are negatively autocorrelated ($I = -0.105$), because the conditional mode borrows strength from the neighbours and over-smooths; such residuals cannot be used to test the adequacy of the latent SAR. The predicted innovations $\hat{\mathbf{u}} = (\mathbf{I} - \hat{\rho}\mathbf{W})\hat{\boldsymbol{\eta}} - \mathbf{X}\hat{\boldsymbol{\beta}}$ retain a positive autocorrelation ($I = 0.19$, against 0.22 for the random effects of the ZINB-RE), suggesting that part of the dependence operates at scales that first-order contiguity does not capture.

Spatial weights. With 6- and 10-nearest-neighbour matrices (Appendix B), $\hat{\rho}$ is 0.584 and 0.633, the inflation coefficients and the signs of all count coefficients are unchanged, and the 10-nearest-neighbour matrix fits slightly better than queen contiguity (AIC lower by 135), in line with the residual dependence above.

Cumulative adoption. As a robustness check we fit the same specifications to the cumulative number of installations over the whole life of the scheme, April 2010 to March 2019 (mean 21.0, variance 458, 798 zeros, 2.5%). Zero-inflation lowers the AIC of the NB from 241,464 to 240,689, and ZISAR to about 227,860, a gain of almost 13,000 points; the expected number of zeros is 626 for the NB, 928 for the ZINB and 861 for ZISAR, against 798 observed. The spatial and inflation parameters are close to those of the main analysis: $\hat{\rho} = 0.67$, and the structural-zero probability increases with flats ($\hat{\gamma} = 1.68$) and density (0.71) and decreases with private renting (-0.42). With counts in the hundreds the optimiser stops with a false-convergence flag and the Hessian is not positive definite at the stopping point, so we do not report standard errors. Restarting

the optimiser from the stopping point, with κ free or held at its value there, moves the estimates only slightly: across the restarts we tried the log-likelihood changes by less than 20 points, $\hat{\rho}$ stays between 0.655 and 0.670 and the coefficient of flats between 1.68 and 1.92, which leaves the conclusions unchanged.

Interpretation. The estimate of ρ measures how strongly the latent adoption propensities of neighbouring areas move together, given the covariates. Peer effects of the kind documented by Bollinger and Gillingham [2012] and Graziano and Gillingham [2015] generate such dependence, but so do spatially correlated unobservables—the reach of local installers, roof orientation and shading, local campaigns—and a cross-section cannot separate the two [Manski, 1993, Gibbons and Overman, 2012]. What the application establishes is narrower and, for inference on the covariates, sufficient: treating post-reform adoption as spatially independent, or ignoring the structural zeros, distorts the estimated role of tenure, age and housing form, and leaves strong dependence in the residuals.

Computation. A ZISAR fit, including the Hessian and the standard errors, takes about three minutes on one core of a laptop (Apple M5 Pro), with the log-determinant computed by sparse Cholesky factorisation; the restricted fits take between one and four minutes, and the effects decomposition with $B = 500$ draws about six. The dense eigendecomposition of $\mathbf{W}$, by contrast, would require some 8 GB of memory for the matrix alone.

7 Conclusion

This paper introduced the ZISAR (Zero-Inflated Spatial Autoregressive) model, a unified specification for count data that simultaneously addresses spatial spillovers, zero-inflation, overdispersion, and unobserved unit-level heterogeneity. The key innovation is the placement of a simultaneous autoregressive structure on the latent linear predictor of the count process—$\boldsymbol{\eta} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{u})$—rather than on the observed count. This SAR-on-$\boldsymbol{\eta}$ formulation preserves the direct/indirect/total effects decomposition that is central to the spatial econometrics programme [LeSage and Pace, 2009, Arbia, 2006] while avoiding the endogeneity of observed spatial lags that complicates SAR-on-$\mathbf{y}$ specifications [Lambert et al., 2010]. Simultaneously, it differs from CAR-based spatial random-effects models [Lee, 2013, Neelon et al., 2013] by providing a structural, single-parameter characterisation of spatial dependence with an explicit spillover interpretation.

Estimation by Laplace-approximated marginal likelihood, implemented via `TMB` [Kris-

tensen et al., 2016], yields a computationally efficient and scalable algorithm. The sparse-matrix operations and automatic differentiation exploited by `TMB` echo the computational concerns raised by Arbia and Nardelli [2024] and Arbia et al. [2022] in the context of pairwise pseudo-likelihood for large spatial datasets. We established consistency and asymptotic normality of the estimator and derived the effects decomposition for the nonlinear zero-inflated mixture.

The Monte Carlo study confirmed that ignoring either spatial dependence or zero-inflation leads to substantial bias: the standard ZINB inflates covariate effects when $\rho > 0$, while the SAR-NB severely underestimates the spatial lag when structural zeros are present, with a bias that does not vanish as n grows. ZISAR recovers the data-generating parameters with low bias and an RMSE that decreases with n. A second experiment showed that residual-based Moran tests can miss spatial dependence that the likelihood detects.

The application to domestic solar PV adoption in England after the 2016 reform of the Feed-in Tariff took the estimator to $n = 32{,}322$ small areas, where a fit takes about three minutes. ZISAR found strong spatial dependence in the latent propensity to adopt ($\hat{\rho} = 0.63$), attributed about one tenth of the 10,442 zeros to a structural regime confined to flat-dominated urban cores—a classification that agrees with the areas' adoption history before the reform, which the model does not use—and dominated every competitor in AIC and BIC. It also showed what is lost by ignoring space: the aspatial models credit social renting and the elderly population with effects that disappear once the spatial clustering of adoption is modelled. The application also brought out a practical point for zero-inflated models of administrative counts: a mismatch between the geography of the register and that of the covariates creates artificial zeros, which must be removed before the zero process can be interpreted.

Limitations. Three limitations warrant acknowledgement. First, the Laplace approximation may lose accuracy when the conditional posterior of $\boldsymbol{\eta}$ is highly non-Gaussian (e.g. for very sparse data with many zero counts); higher-order Laplace corrections or adaptive quadrature could improve precision in such settings. Second, the unit-level random effects $\mathbf{u}$ are assumed independent Gaussian; correlated or non-Gaussian specifications would add flexibility at the cost of a more complex `TMB` template. Third, the eigenvalue pre-computation for the log-determinant $\log|\det(\mathbf{I} - \rho\mathbf{W})|$ is $O(n^3)$; the sparse-Cholesky alternative of (22) removes this bottleneck at the national scale of the application, but its cost grows with the fill-in of the Cholesky factor, so that for dense weights matrices or n in the millions stochastic log-determinant approximations [e.g. Barry and Pace, 1999] would be needed.

Extensions. Several directions for future research emerge naturally. The most immediate is the incorporation of *temporal dependence*, yielding a spatio-temporal ZISAR of the form

$$\eta_{it} = \rho(\mathbf{W}\boldsymbol{\eta}_t)_i + \phi\,\eta_{i,t-1} + \mathbf{x}_{it}'\boldsymbol{\beta} + u_i + v_t,$$

where ϕ captures persistence and v_t is a time-specific effect. The panel structure introduces additional identification and computational challenges (dynamic spatial panel asymptotics, high-dimensional latent field) that merit a dedicated treatment.

A second extension concerns alternative spatial structures: replacing the single $\mathbf{W}$ with multiple weight matrices (e.g. geographic and economic proximity), or letting $\mathbf{W}$ be endogenous.

Third, the ZISAR template naturally accommodates other count distributions (generalised Poisson, Conway–Maxwell–Poisson) and alternative inflation links (probit, complementary log-log) by modifying the observation-model block of the `TMB` template.

Finally, packaging the estimator as a fully documented `R` package with formula interfaces, spatial-weights integration (`spdep`/`sf`), and automated model-selection tools would lower the barrier to adoption and is planned for a subsequent release.

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A Proofs

A.1 Proof of Proposition 1

We verify the conditions of the extremum-estimator consistency theorem [Newey and McFadden, 1994, Theorem 2.1].

Laplace approximation error. Let $\ell(\boldsymbol{\theta}) = \log \int p(\mathbf{y}, \boldsymbol{\eta} \mid \boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\eta}$ denote the exact marginal log-likelihood and $\ell_{\mathrm{LA}}(\boldsymbol{\theta})$ the Laplace approximation (20). By Tierney and Kadane [1986], $|\ell_{\mathrm{LA}}(\boldsymbol{\theta}) - \ell(\boldsymbol{\theta})| = O(1)$ uniformly over Θ (the error is bounded because the conditional density $f(y_i \mid \eta_i, \pi_i; \kappa)$ has bounded derivatives in η_i under Assumption 3). Dividing by n: $|n^{-1}\ell_{\mathrm{LA}}(\boldsymbol{\theta}) - n^{-1}\ell(\boldsymbol{\theta})| = O(n^{-1}) \to 0$.

Uniform convergence. The exact marginal log-likelihood satisfies $n^{-1}\ell(\boldsymbol{\theta}) \to Q(\boldsymbol{\theta}) \equiv \mathbb{E}[\log p(y_1 \mid \boldsymbol{\theta})]$ uniformly over Θ, by the uniform law of large numbers. The key requirement is that the summands are bounded and equicontinuous in $\boldsymbol{\theta}$, which holds under the compactness of Θ (Assumption 3) and the boundedness of $\mathbf{W}$ (Assumption 1). Combined with the vanishing Laplace error, $\sup_{\boldsymbol{\theta}\in\Theta} |n^{-1}\ell_{\mathrm{LA}}(\boldsymbol{\theta}) - Q(\boldsymbol{\theta})| \xrightarrow{p} 0$.

Identification. By Assumption 4, $Q(\boldsymbol{\theta})$ has a unique maximum at $\boldsymbol{\theta}_0$.

Conclusion. Compactness of Θ, uniform convergence, and identifiable uniqueness satisfy the conditions of the extremum-estimator theorem, giving $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$. □

A.2 Proof of Proposition 2

Score expansion. By a mean-value expansion around $\boldsymbol{\theta}_0$:

$$\mathbf{0} = n^{-1/2}\frac{\partial \ell_{\mathrm{LA}}}{\partial \boldsymbol{\theta}}\bigg|_{\hat{\boldsymbol{\theta}}} = n^{-1/2}\frac{\partial \ell_{\mathrm{LA}}}{\partial \boldsymbol{\theta}}\bigg|_{\boldsymbol{\theta}_0} + n^{-1}\frac{\partial^2 \ell_{\mathrm{LA}}}{\partial \boldsymbol{\theta}\,\partial \boldsymbol{\theta}'}\bigg|_{\bar{\boldsymbol{\theta}}} \sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0).$$

CLT for the score. The Laplace-approximate score $\partial \ell_{\mathrm{LA}}/\partial \boldsymbol{\theta}$ can be decomposed as a sum of local contributions $s_i(\boldsymbol{\theta})$ that depend on unit i and its spatial neighbours. Under Assumption 1, the spatial weights have uniformly bounded row and column sums, so each s_i depends on at most $O(1)$ other units (the neighbourhood is bounded). This induces a mixing structure that satisfies the CLT for spatially dependent arrays [see Lee, 2004, Theorem 3.1]:

$$n^{-1/2}\sum_{i=1}^{n} s_i(\boldsymbol{\theta}_0) \xrightarrow{d} N\big(\mathbf{0},\, \boldsymbol{\Omega}_0\big),$$

where $\boldsymbol{\Omega}_0 = \lim_{n\to\infty} \mathrm{Var}\left[n^{-1/2}\sum_i s_i(\boldsymbol{\theta}_0)\right]$.

Hessian convergence. By the uniform LLN,

$$n^{-1}\frac{\partial^2 \ell_{\mathrm{LA}}}{\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}'}\bigg|_{\tilde{\boldsymbol{\theta}}} \xrightarrow{p} \mathcal{H}_0 \equiv \mathbb{E}\left[\frac{\partial^2 \log p(y_1 \mid \boldsymbol{\theta})}{\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}'}\bigg|_{\boldsymbol{\theta}_0}\right],$$

which is negative definite by Assumption 4. Under standard regularity (information matrix equality), $\mathcal{H}_0 = -\mathcal{I}(\boldsymbol{\theta}_0) = -\boldsymbol{\Omega}_0$.

Combining. Rearranging the score expansion:

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\mathcal{H}_0^{-1}\, n^{-1/2}\sum_i s_i(\boldsymbol{\theta}_0) + o_p(1) \xrightarrow{d} N\big(\mathbf{0},\, \mathcal{I}(\boldsymbol{\theta}_0)^{-1}\big).$$

□

A.3 Proof of Proposition 3

Under the restricted ZISAR ($\mathbf{g} = \mathbf{0}$), the marginal expectation of y_i evaluated at the conditional mode is

$$\mathbb{E}[y_i] \approx (1-\pi_i)\,\mu_i, \qquad \mu_i = \exp(\eta_i), \quad \pi_i = \mathrm{logit}^{-1}(\mathbf{z}_i'\boldsymbol{\gamma}).$$

The latent predictor at the mode satisfies $\hat{\boldsymbol{\eta}} \approx \mathbf{S}(\rho)\mathbf{X}\boldsymbol{\beta}$ (ignoring the random-effect contribution, which integrates out at the mode). The partial derivative of $\boldsymbol{\eta}$ with respect to x_{jr} is

$$\frac{\partial \eta_i}{\partial x_{jr}} = [\mathbf{S}(\rho)]_{ij}\,\beta_r.$$

The effect on $\mathbb{E}[y_i]$ is

$$\frac{\partial \mathbb{E}[y_i]}{\partial x_{jr}} = (1-\pi_i)\,\mu_i\,[\mathbf{S}(\rho)]_{ij}\,\beta_r + \underbrace{(\text{inflation channel})}_{\text{diagonal only}}.$$

Stacking: $\mathbf{M}_r = \mathrm{diag}((1-\hat{\boldsymbol{\pi}}) \odot \hat{\boldsymbol{\mu}})\,\mathbf{S}(\rho)\,\beta_r$.

(i) When $\rho \in [0,1)$ and $\mathbf{W} \geq 0$ entrywise, the power-series $\mathbf{S}(\rho) = \sum_{s=0}^{\infty}\rho^s\mathbf{W}^s$ has all non-negative entries. Since $(1-\pi_i)\mu_i > 0$, $\mathbf{M}_r$ has the same sign as β_r.

(ii) When $\rho = 0$, $\mathbf{S}(\rho) = \mathbf{I}_n$ and $\mathbf{M}_r$ is diagonal, so all indirect effects (off-diagonal elements) are zero. For $\rho \neq 0$, $\mathbf{S}(\rho)$ has positive off-diagonal elements (since $\mathbf{W}^s$ has positive entries for $s \geq 1$ on connected lattices), making indirect effects non-zero.

(iii) Each element of $\mathbf{S}(\rho) = \sum_{s=0}^{\infty} \rho^s \mathbf{W}^s$ is monotonically increasing in $|\rho|$ for $\rho \geq 0$ (since all terms $\rho^s [\mathbf{W}^s]_{ij}$ are non-negative and increasing in ρ). □

B Sensitivity Analysis

Table 11 reports the ZISAR estimates of the application (Section 6) under alternative spatial weights: the row-standardised queen-contiguity matrix of the main specification and the 6- and 10-nearest-neighbour matrices built on the population-weighted centroids of the LSOAs. The spatial parameter, the inflation coefficients and the signs of all count coefficients are stable; the 10-nearest-neighbour matrix has the lowest AIC, and the shape parameter κ is at the upper bound of the optimisation box in all three fits.

Table 11: Sensitivity of the ZISAR estimates to the choice of spatial weights ($n = 32{,}322$). Standard errors of $\hat{\rho}$ and $\hat{\sigma}$ in parentheses.

	Queen	k-NN ($k = 6$)	k-NN ($k = 10$)
$\hat{\rho}$	0.631 (0.011)	0.584 (0.009)	0.633 (0.009)
$\hat{\sigma}$	0.554 (0.008)	0.585 (0.006)	0.587 (0.006)
Count process			
Households (log)	0.150	0.164	0.160
Detached	0.093	0.107	0.103
Social rented	−0.020	−0.008	−0.023
Income deprivation	0.037	0.030	0.045
Degree	0.032	0.030	0.041
Aged 65+	−0.006	−0.003	−0.013
Off gas	0.043	0.050	0.053
Northing	−0.025	−0.036	−0.031
Flats	−0.097	−0.121	−0.103
Density (log)	−0.088	−0.133	−0.128
Private rented	−0.048	−0.038	−0.052
Inflation process			
Flats	1.993	2.321	2.132
Density (log)	0.868	0.823	0.822
Private rented	−0.289	−0.282	−0.273
Log-likelihood	−53082	−53134	−53014
AIC	106201	106307	106066

C TMB Implementation Details

The `TMB` template declares $\boldsymbol{\eta}$ as random effects and uses the SAR prior density:

$$-\log p(\boldsymbol{\eta} \mid \boldsymbol{\beta}, \rho, \sigma) = \frac{n}{2}\log(2\pi\sigma^2) - \log|\det(\mathbf{I} - \rho\mathbf{W})| + \frac{1}{2\sigma^2}\|(\mathbf{I} - \rho\mathbf{W})\boldsymbol{\eta} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

The log-determinant is computed via pre-computed eigenvalues $\lambda_1, \dots, \lambda_n$ of $\mathbf{W}$ (passed as data). For possibly complex eigenvalues ($\lambda_j = a_j + b_j \mathrm{i}$):

$$\log \left| \det(\mathbf{I} - \rho \mathbf{W}) \right| = \frac{1}{2} \sum_{j=1}^{n} \log \left[(1 - \rho a_j)^2 + (\rho b_j)^2 \right].$$

For large n (the default in `zisar_fit` when $n > 5000$, or with `logdet = "sparse"`), the template receives instead the sparse matrices $\mathbf{I}_n$, $\mathbf{W} + \mathbf{W}'$ and $\mathbf{W}'\mathbf{W}$, assembles $\mathbf{Q}(\rho) = \mathbf{I}_n - \rho(\mathbf{W} + \mathbf{W}') + \rho^2 \mathbf{W}'\mathbf{W}$ and evaluates (22) as $\frac{1}{2}$`newton::log_determinant`$(\mathbf{Q})$, an atomic function of `TMB`'s `TMBad` framework that factorises $\mathbf{Q}$ by sparse Cholesky and supplies its derivative with respect to ρ. The same mechanism is used by `TMB` for Gaussian Markov random fields with parameter-dependent precision. The sparse matrix–vector product $(\mathbf{I} - \rho \mathbf{W})\boldsymbol{\eta}$ is computed via Eigen's sparse algebra, preserving the $O(\mathrm{nnz}(\mathbf{W}))$ complexity. The reparametrisation $\rho = \tanh(\tilde{\rho})$ ensures $\rho \in (-1, 1)$; for numerical stability the optimiser additionally bounds $\tilde{\rho} \in [-4, 4]$, $\kappa \in [0.01, 500]$ and $\sigma \in [0.001, 10]$ (Section 4.2). Standard errors on the original scale are obtained by the delta method through `TMB`'s `ADREPORT` mechanism.